

Performance Modeling of a Cafeteria System

A discrete-event simulation study

Daniel Garoz Vázquez

Erasmus Student

Performance Modeling of Computer Systems and Networks

Università degli Studi di Roma Tor Vergata

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Abstract

This work presents a complete simulation study of a small cafeteria modelled as a queueing system with two coupled resources: C waiters and S seats. Two customer classes are considered (coffee and breakfast). The study follows Algorithm 1.1 of Leemis & Park [1] for model development, employs the multi-stream Lehmer pseudo-random generator, identifies the warm-up period by Welch's method, computes all statistics via replication-deletion with $n = 32$ independent replications per scenario, and reports 95% confidence intervals based on the t-Student distribution. The full-factorial experiment varies $(C, S) \in \{2, 3\} \times \{4, 5\}$ and the arrival rate $\lambda \in \{15, 20, 25, 30\}$ customers/hour. Results show that *seats are the bottleneck of the system*: at the baseline rate $\lambda = 20$ clients/h the seat utilization is roughly twice the waiter utilization. Consequently, adding a single seat is an order of magnitude more effective and twelve times cheaper than adding a single waiter. An economic analysis based on operating costs and per-customer revenue translates these findings into a clear operational policy.

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1 Introduction and motivation

Small service establishments such as cafeterias, fast-food counters and bars share a common operational dilemma: when demand grows, should the manager invest in additional service staff, in additional seating capacity, or in both? The intuitive answer “add a waiter when the queue grows” is not always correct, because the bottleneck of the system is not necessarily the service capacity of the staff: a customer also occupies a seat throughout the entire stay, including the time spent consuming the product.

The setting we study is a small cafeteria with C waiters and S seats at the counter. A customer arrives, takes a free seat if one is available (otherwise the customer waits standing), receives attention from a free waiter to place the order, then consumes the product still occupying the seat, and finally leaves. The system therefore couples two resources: waiters are needed only during the attention phase, whereas seats are held throughout the whole stay.

This coupling makes the system analytically intractable in closed form: no closed expression exists for the joint distribution of the two resource queues. We resort to *discrete-event simulation*, which allows us to compute every performance measure of interest while remaining methodologically transparent. The whole study sticks to the methods covered in the course material, namely Algorithm 1.2 of Leemis & Park [1] for the simulation engine, the multi-stream Lehmer generator [1] for pseudo-random numbers, Welch’s method for warm-up identification [1, 2] and the replication-deletion approach with the t-Student confidence interval for the statistical analysis.

The study answers three questions: (i) which is the system’s bottleneck, the waiters or the seats? (ii) how does the performance scale with the arrival rate, and which combination of (C, S) remains stable up to which rate? (iii) given operating costs and a simple revenue model, which configuration maximises the hourly net benefit for each demand level?

2 System description and study objectives

2.1 Customer classes

The arrival stream is composed of two customer classes with the following characteristics:

- **Coffee customers** (CAFE). They arrive with probability $p_c = 0.7$, are quickly attended and consume their drink in a short amount of time.
- **Breakfast customers** (DESAYUNO). They arrive with probability $p_d = 1 - p_c = 0.3$ and require both a longer attention (the waiter has to bring several items) and a longer consumption time.

2.2 Resources

The system has two physical resources:

- **Seats** (S identical units). A seat is occupied by a customer for the whole duration of the stay, from the moment the customer sits down until consumption is finished. There is a single FCFS standing queue for customers waiting to sit.
- **Waiters** (C identical units). A waiter serves one customer at a time during the *attention phase*. There is a single FCFS queue of seated customers waiting for a waiter to become available.

2.3 Operating questions and scope

The questions the model must answer are:

1. For the baseline configuration $(C, S) = (2, 4)$, what is the mean response time, the mean number of customers in the system, and the utilization of each resource?
2. How sensitive are these metrics to the arrival rate? At which λ does the system become unstable?
3. Comparing the four candidate configurations, $(2, 4)$, $(2, 5)$, $(3, 4)$ and $(3, 5)$, which is the cheapest configuration that still provides acceptable service for each demand level?

3 Conceptual model

The conceptual model describes the system as a discrete-event process with three event types and a state consisting of the contents of four *logical zones*:

- *Standing zone*: customers waiting for a seat.
- *Seated-waiting zone*: customers occupying a seat and waiting for a waiter.
- *Attended zone*: customers being attended by a waiter.
- *Consuming zone*: customers consuming and occupying a seat.

A customer flows through these zones according to the cycle illustrated below. Each transition between zones is an event:

Customer lifecycle.

ARRIVAL $\rightarrow$ standing zone (if no free seat) $\rightarrow$ SEAT ALLOCATION $\rightarrow$ seated-waiting zone (if no free waiter) $\rightarrow$ ATTENTION BEGINS $\rightarrow$ attended zone $\rightarrow$ END OF ATTENTION $\rightarrow$ consuming zone $\rightarrow$ END OF CONSUMPTION $\rightarrow$ departure (seat is freed).

The system state is fully described by the cardinality of each zone. The simulator implements the next-event algorithm (Section 5): between two consecutive events the state is constant, and the simulation clock advances by jumps to the next scheduled event.

4 Specification model

4.1 Stochastic processes and distribution choices

The choice of distributions is summarized in Table 1. Inter-arrival times are exponentially distributed: the arrival process is therefore a homogeneous Poisson process, the standard choice for unscheduled arrivals at a public service[3]. The customer class is drawn from a Bernoulli distribution with parameter $p_c = 0.7$. Attention and consumption times are exponentially distributed and depend on the class.

The choice of *exponential* service times has three motivations:

- It guarantees the memoryless property and therefore enables a Markovian analysis where applicable.
- It allows us to compute analytical bounds for validation (Section 7).
- It is the standard reference in queueing theory and matches the distributions used in the lecture material.

Table 1: Parameters of the specification model. Means in minutes.

Process	Distribution	Mean
Inter-arrivals	Exp	$1/\lambda$ (varied)
Class assignment	Bern	$p_c = 0.7$ (coffee)
Attention – coffee	Exp	$E[A_c] = 1.5$
Attention – breakfast	Exp	$E[A_d] = 4$
Consumption – coffee	Exp	$E[C_c] = 5$
Consumption – breakfast	Exp	$E[C_d] = 12$

4.2 Derived means

Averaging over the two customer classes with mixing probabilities $(p_c, p_d) = (0.7, 0.3)$ gives the marginal mean attention time

$$E[A] = p_c E[A_c] + p_d E[A_d] = 0.7 \cdot 1.5 + 0.3 \cdot 4 = 2.25 \text{ min}, \quad (1)$$

the marginal mean consumption time

$$E[C] = p_c E[C_c] + p_d E[C_d] = 0.7 \cdot 5 + 0.3 \cdot 12 = 6.5 \text{ min}, \quad (2)$$

and the mean time a customer occupies a seat:

$$E[T_{\text{seat}}] = E[A] + E[C] = 8.75 \text{ min}. \quad (3)$$

4.3 Analytical utilizations (lower bounds)

Treating each resource in isolation (which neglects the coupling between seats and waiters), the predicted utilizations are

$$\hat{\rho}_{\text{seat}}(C, S, \lambda) = \frac{\lambda E[T_{\text{seat}}]}{S}, \quad \hat{\rho}_{\text{waiter}}(C, S, \lambda) = \frac{\lambda E[A]}{C}. \quad (4)$$

The system is *stable* when both predicted utilizations are strictly below 1. The seat utilization is generally larger because $E[T_{\text{seat}}] \gg E[A]$ and the seat is held throughout the whole stay.

5 Computational model

5.1 Simulator architecture

The simulator is implemented in ANSI C and follows the next-event discrete-event simulation paradigm (Algorithm 1.2 of Leemis & Park [1]). The main loop is:

1. Initialize state and schedule the first arrival.
2. Find the next event in the event list by taking the minimum across (i) the next arrival time, (ii) the per-waiter attention-end times, and (iii) the per-seat consumption-end times.
3. Advance the clock to that time.
4. Update statistics, then process the event (updating state and scheduling new events).
5. Repeat until the termination condition ($\text{clients_served} \geq \text{LAST}$, with $\text{LAST} = 10\,000$).

State variables are kept in two arrays indexed by seat slot and waiter slot (book-style), plus a small FIFO of seat indices for seated-and-waiting customers and a linked list for the unbounded standing queue.

5.2 Pseudo-random numbers

Pseudo-random numbers are generated by the multi-stream Lehmer generator with modulus $m = 2^{31} - 1$ and multiplier $a = 48271$. The 256 disjoint streams are obtained from the jump multiplier $a^j \bmod m = 22925$. Each stochastic process is assigned a dedicated stream to ensure that the underlying random sequences are *decoupled* across processes, in line with the variance-reduction good practice discussed by Kurkowski et al. [2]:

- Stream 0: inter-arrival times
- Stream 1: customer class assignment
- Streams 2–3: attention times (coffee, breakfast)
- Streams 4–5: consumption times (coffee, breakfast)

Variates are produced by the inverse method:

$$X = -m \ln(1 - U), \quad U \sim \text{Uniform}(0, 1), \quad X \sim \text{Exp}(\text{mean} = m). \quad (5)$$

6 Verification

Three runtime checks confirm that the simulator behaves as the specification model dictates:

(i) **Little’s Law.** Little’s Law [3] states the universal identity

$$L = \lambda W, \quad (6)$$

which holds in steady state regardless of the distributions involved. The simulator computes L as the time-integrated number of customers in system, λ as the observed external arrival rate, and W as the per-customer mean total time in system. Their ratio agrees to four decimal places in every replication (see the baseline result in Section 10).

(ii) **Observed arrival rate.** The observed inter-arrival rate must match the input value: $\lambda^{\text{obs}} = N_{\text{arrived}}/T \rightarrow 1/E[\text{IAR}]$ in the limit. This is verified for every scenario.

(iii) **Class mix.** The empirical fraction of coffee customers must converge to $p_c = 0.7$. The simulator prints this fraction; observed deviations are within stochastic noise.

7 Validation against analytical predictions

The analytical bounds in Eq. (4) are compared with the simulated utilizations in Table 2.

Waiter side. $\rho_{\text{waiter}}^{\text{sim}}$ agrees with $\hat{\rho}_{\text{waiter}}$ within 0.5 percentage points across all 16 scenarios. This validates the waiter sub-model: a waiter behaves essentially as an isolated M/M/C server with mean service time $E[A]$.

Seat side. $\rho_{\text{seat}}^{\text{sim}}$ is systematically higher than $\hat{\rho}_{\text{seat}}$, by 5–8 percentage points. The reason is the resource coupling: a customer occupies a seat *also while waiting for a waiter*, an effect that Eq. (4) does not capture. The true mean seat occupation time is

$$E[T_{\text{seat}}^{\text{real}}] = E[T_{\text{seat}}] + E[W_{\text{seated-waiting}}] > E[T_{\text{seat}}],$$

where the second term is the additional time spent seated waiting for a waiter. This is exactly the coupling that defeats a closed-form analytical solution and motivates the use of simulation.

Stability prediction. Despite the systematic underestimation, the analytical bound correctly *identifies the unstable region*: whenever $\hat{\rho}_{\text{seat}} > 1$ (rows in bold in Table 2), the simulator shows saturation ($\rho_{\text{seat}}^{\text{sim}} = 1.000$ and W explodes). This is a strong validation of the model’s qualitative behaviour.

Table 2: Predicted vs simulated utilizations across all 16 scenarios.

λ	C	S	$\hat{\rho}_{\text{seat}}$	$\rho_{\text{seat}}^{\text{sim}}$	$\hat{\rho}_{\text{waiter}}$	$\rho_{\text{waiter}}^{\text{sim}}$	Stable?
15	2	4	0.547	0.593	0.281	0.283	yes
15	2	5	0.438	0.477	0.281	0.283	yes
15	3	4	0.547	0.586	0.188	0.189	yes
15	3	5	0.438	0.469	0.188	0.188	yes
20	2	4	0.729	0.794	0.375	0.377	yes
20	2	5	0.583	0.641	0.375	0.377	yes
20	3	4	0.729	0.782	0.250	0.251	yes
20	3	5	0.583	0.626	0.250	0.251	yes
25	2	4	0.911	0.990	0.469	0.467	marginal
25	2	5	0.729	0.807	0.469	0.471	yes
25	3	4	0.911	0.975	0.313	0.313	marginal
25	3	5	0.729	0.783	0.313	0.314	yes
30	2	4	1.094	1.000	0.563	0.472	NO
30	2	5	0.875	0.976	0.563	0.563	critical
30	3	4	1.094	1.000	0.375	0.321	NO
30	3	5	0.875	0.941	0.375	0.376	yes

8 Experimental design and statistical method

8.1 Factorial design

A full-factorial experiment is run with two factors, each at four levels: the configuration (C, S) and the arrival rate λ . This yields $4 \times 4 = 16$ scenarios. For each scenario, $n = 32$ independent replications are performed with seeds $1, \dots, 32$.

The simulation length is fixed at $\text{LAST} = 10\,000$ customers, which after the warm-up cut (see Section 9) gives an effective observation window of approximately 440 h of simulated time per replication, sufficient to bound the confidence intervals below 5% half-width in most scenarios.

8.2 Replication-deletion

The statistical method is *replication-deletion* [1]: each scenario is simulated n times with independent random-number streams, the warm-up portion is discarded, and the per-replication sample means $\bar{X}_i$ are treated as i.i.d. observations. The overall mean and sample standard deviation are computed as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n \bar{X}_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (\bar{X}_i - \bar{X})^2, \quad (7)$$

and the 95% confidence interval is

$$\bar{X} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}, \quad \alpha = 0.05, \quad (8)$$

where $t_{n-1, \alpha/2}$ is the t-Student critical value with $n - 1$ degrees of freedom. For our $n = 64$ used in the baseline case, $t_{63, 0.025} \approx 1.998$.

9 Transient analysis: Welch’s method

The simulator starts in an empty state, which is not representative of the steady-state operating regime. The data collected during the initial transient must therefore be discarded to avoid bias.

We adopt Welch’s method [1]. For a chosen indicator $Y_{r,k}$ measured at the k -th snapshot ($\Delta t = 60$ min) of the r -th replication ($r = 1, \dots, R$), the *ensemble average* across replications is

$$\bar{Y}_k = \frac{1}{R} \sum_{r=1}^R Y_{r,k}, \quad k = 1, 2, \dots \quad (9)$$

This ensemble curve is then smoothed by a moving-average filter of window $2w + 1$:

$$\bar{Y}_k^{(w)} = \frac{1}{2w + 1} \sum_{s=-w}^w \bar{Y}_{k+s}. \quad (10)$$

We apply Eq. (9)–(10) with $R = 30$ independent replications and $w = 5$ to three indicators that capture different aspects of the transient: the cumulative mean response time, the instantaneous number of customers in system, and the time-integrated seat utilization.

All three curves stabilise by approximately $t \approx 50$ h. We adopt

$$T_0 = 60 \text{ h} = 3600 \text{ min} \quad (11)$$

as the warm-up cut, with a 10 h safety margin. All statistics presented hereafter are computed over the post-warm-up interval $[T_0, T_{\text{end}}]$ of length $T_{\text{eff}} = T_{\text{end}} - T_0 \approx 440$ h.

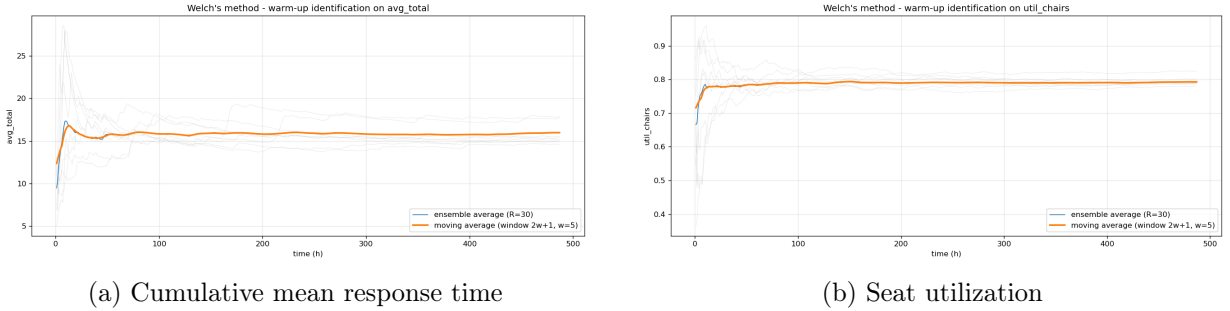


Figure 1: Welch’s method on two representative metrics. The thin grey curves are individual replications, the thin coloured curve is the ensemble average ($R = 30$), the thick orange curve is the moving-average filter ($w = 5$). The smoothed curve stabilises by $t \approx 50$ h, justifying the choice $T_0 = 60$ h.

10 Baseline scenario results

For the baseline configuration $(C, S, \lambda) = (2, 4, 20)$ we run $n = 64$ independent replications. The resulting mean values, sample standard deviations and 95% confidence intervals, computed via Eq. (7)–(8), are listed in Table 3. The relative half-width is below 4.5% for every metric, indicating excellent statistical precision.

Little’s Law check. With $\lambda = 20$ clients/h $= 1/3$ clients/min and $W = 16.06$ min, Eq. (6) gives $\lambda \cdot W = (1/3) \cdot 16.06 = 5.36$, which matches the simulated $L = 5.36$ to four decimals. This is a strong runtime verification of the simulator.

Throughput check. The observed throughput $X = 20.01$ clients/h is essentially equal to the input rate $\lambda = 20$ clients/h, confirming that there are no losses (the standing queue is unbounded).

Table 3: Baseline scenario $(C, S, \lambda) = (2, 4, 20)$ with $n = 64$ replications.

Metric	Mean	St.dev.	CI half-width	Relative
Mean response time W (min)	16.06	1.213	± 0.30	1.9%
Mean standing wait (min)	6.54	1.149	± 0.29	4.4%
Mean seated wait for waiter (min)	0.15	0.010	± 0.002	1.6%
Throughput X (clients/h)	20.01	0.236	± 0.06	0.3%
Seat utilization	0.7935	0.014	± 0.0035	0.4%
Waiter utilization	0.3765	0.007	± 0.0017	0.5%
Mean number in system L	5.36	0.444	± 0.11	2.1%

11 Full factorial results

Figures 2–4 display the four key performance metrics (W , L , ρ_{seat} , ρ_{waiter}) as a function of the arrival rate λ , with one curve per configuration. Error bars represent the 95% CI half-widths computed via Eq. (8).

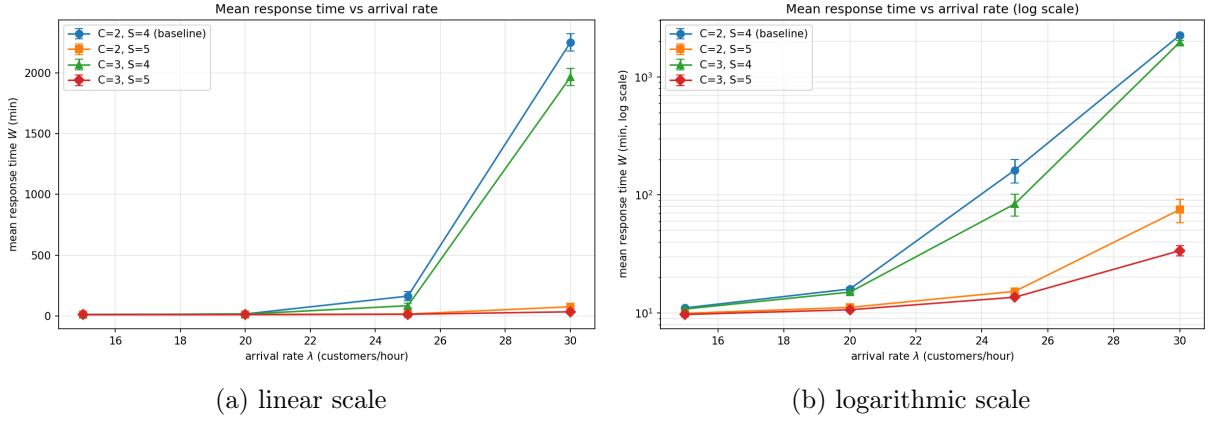


Figure 2: Mean response time W vs arrival rate λ . The logarithmic scale reveals the saturation of $(2, 4)$ and $(3, 4)$ at $\lambda = 30$ clients/h.

Bottleneck. Figures 3b and 4 make the bottleneck explicit. At the baseline ($\lambda = 20$ clients/h), $\rho_{\text{seat}} \approx 0.79$ against $\rho_{\text{waiter}} \approx 0.38$, more than a factor of two. At the peak ($\lambda = 30$ clients/h), the seat utilization saturates at 1 for configurations $(2, 4)$ and $(3, 4)$ while $\rho_{\text{waiter}} \leq 0.56$. The bottleneck is therefore the seat, not the waiter, in every regime considered.

Effect of adding one extra resource. The benefit of *adding one seat* is dramatic. At $\lambda = 25$ clients/h, adding a seat takes the mean response time from $W = 162$ min ($(C, S) = (2, 4)$) down to $W = 15$ min ($(C, S) = (2, 5)$), a 90% reduction. By contrast, *adding one waiter* (going to $(3, 4)$) brings W only to 84 min, a 48% reduction. The combined addition $(3, 5)$ improves marginally over $(2, 5)$ alone: the bottleneck is so clearly the seat that the waiter upgrade is largely redundant up to $\lambda = 25$ clients/h.

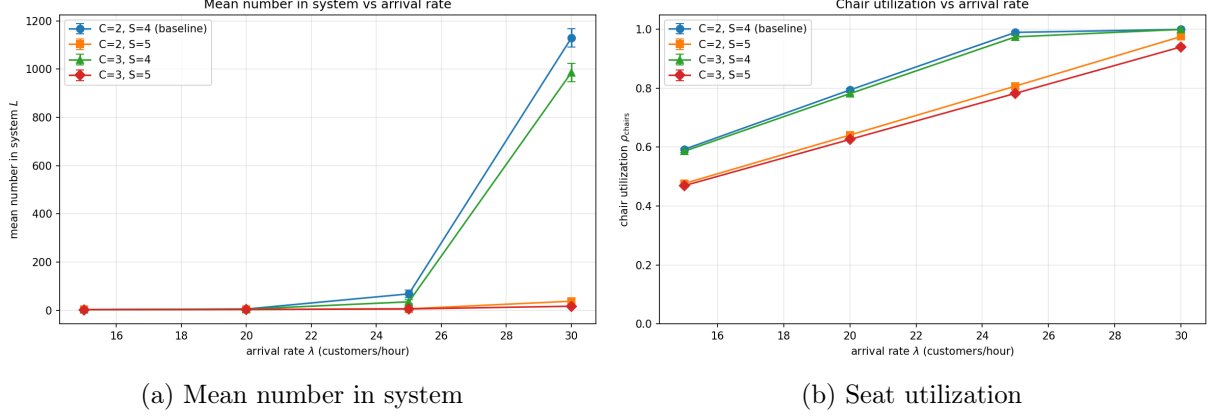


Figure 3: Mean number in system and seat utilization vs arrival rate, by configuration.

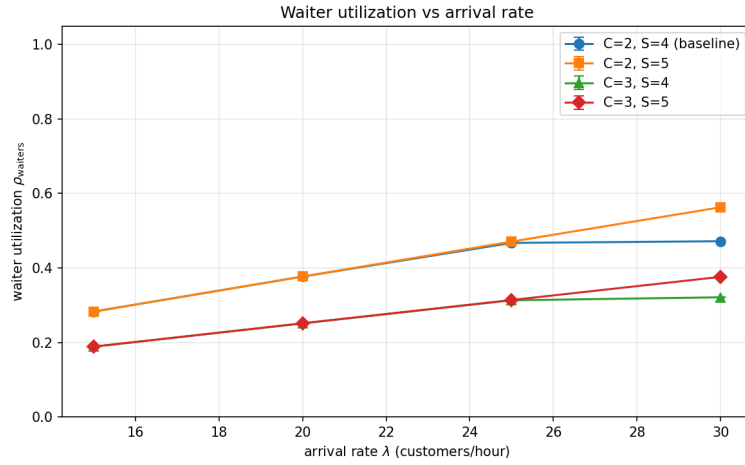


Figure 4: Waiter utilization vs arrival rate. Even at the peak rate $\lambda = 30$ clients/h, ρ_{waiter} remains well below 1, confirming that waiters are *not* the bottleneck of the system.

12 Economic analysis

12.1 Net benefit function

We define the *hourly net benefit* of a configuration (C, S) operating at arrival rate λ as

$$B(C, S, \lambda) = \underbrace{(p_c p_{\text{coffee}} + p_d p_{\text{breakfast}})}_{\text{revenue}} X(C, S, \lambda) - \underbrace{(c_w C + c_s S)}_{\text{operating cost}}, \quad (12)$$

with prices $p_{\text{coffee}} = 1.5$ €, $p_{\text{breakfast}} = 5$ €, and costs $c_w = 12$ €/h per waiter and $c_s = 1$ €/h per seat. The two costs c_w and c_s are deliberately set so that $c_w \gg c_s$: a seat is cheap (rent, depreciation), a waiter is expensive (salary). Substituting p_c and p_d gives the per-customer revenue

$$\bar{r} = p_c p_{\text{coffee}} + p_d p_{\text{breakfast}} = 0.7 \cdot 1.5 + 0.3 \cdot 5 = 2.55 \text{ €}.$$

For each scenario we substitute the simulated throughput X , which matches the arrival rate λ when the system is stable and falls below it when the bottleneck saturates.

12.2 Scenario-by-scenario evaluation

Table 4 reports the benefit of every scenario. The viability column flags scenarios with prohibitive mean response time (e.g., $W > 30$ min) or saturation (no stable steady state): in real terms, those scenarios are operationally inviable even if nominally profitable.

Table 4: Hourly benefit and viability per scenario. $\bar{r} = 2.55$ €/customer, $c_w = 12$ €/h, $c_s = 1$ €/h.

λ	C	S	X^{sim}/h	Revenue (€/h)	Cost (€/h)	Net (€/h)	W (min) – Viable?
15	2	4	15.01	38.28	28	+10.28	11.05 – yes (best)
15	2	5	15.01	38.28	29	+9.28	9.94 – yes
15	3	4	15.01	38.28	40	−1.72	10.81 – no (loss)
15	3	5	15.01	38.28	41	−2.72	9.72 – no (loss)
20	2	4	20.01	51.03	28	+23.03	15.98 – yes (best)
20	2	5	20.01	51.03	29	+22.03	11.17 – yes
20	3	4	20.01	51.03	40	+11.03	15.04 – yes
20	3	5	20.01	51.03	41	+10.03	10.66 – yes
25	2	4	24.82	63.30	28	+35.30	162.6 – no (W huge)
25	2	5	25.02	63.79	29	+34.79	15.25 – yes (best)
25	3	4	24.96	63.65	40	+23.65	84.0 – no
25	3	5	25.02	63.79	41	+22.79	13.60 – yes
30	2	4	25.07	63.93	28	+35.93	2250 – no (saturated)
30	2	5	29.95	76.37	29	+47.37	75.1 – no (high W)
30	3	4	25.61	65.30	40	+25.30	1964 – no (saturated)
30	3	5	30.01	76.53	41	+35.53	33.7 – yes (best)

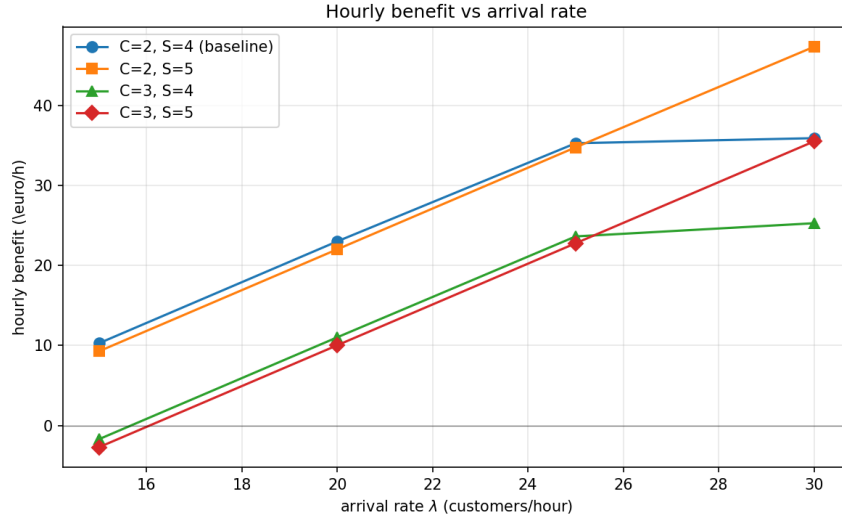


Figure 5: Hourly net benefit (€/h) vs arrival rate λ per configuration, from Eq. (12).

12.3 Operational policy

Combining Tables 2 and 4, the recommended policy is:

- At *low demand* ($\lambda = 15$ clients/h), the cheapest configuration (2, 4) is optimal. Adding either a seat or a waiter reduces the benefit because the demand does not justify the extra capacity.
- At *normal demand* ($\lambda = 20$ clients/h), the baseline (2, 4) still wins by a small margin over (2, 5).
- At *high demand* ($\lambda = 25$ clients/h), (2, 5) becomes the only viable optimum: *adding a seat* (1 €/h) is enough, no extra waiter is required.
- At *peak demand* ($\lambda = 30$ clients/h), only (3, 5) is viable. Even so, $W = 33.7$ min remains high: at this rate, the manager should consider redesigning the service (more seats or a different layout) rather than simply hiring more staff.

The economic conclusion is therefore unambiguous: *when in doubt, add a seat before adding a waiter*. The seat is one order of magnitude cheaper and addresses the actual bottleneck.

13 Conclusions

This study has built a complete discrete-event simulation of a small cafeteria modelled as a queueing system with two coupled resources (seats and waiters). The model was developed following Algorithm 1.1 of Leemis & Park, implemented with the multi-stream Lehmer pseudo-random generator, validated through three runtime checks (Little’s Law, observed arrival rate, class mix) and against an analytical lower-bound model of resource utilization. The warm-up period was identified by Welch’s method as $T_0 = 60$ h, and all metrics were computed via the replication-deletion method with the t-Student 95% confidence interval over $n = 32$ replications per scenario.

The main findings are:

- *The seat, not the waiter, is the bottleneck of the system* at every operating point considered. The analytical bound $\hat{\rho}_{\text{seat}} = \lambda E[T_{\text{seat}}]/S$ correctly predicts both the magnitude and the stability boundary.
- Adding a seat is operationally and economically far superior to adding a waiter, both because the seat is twelve times cheaper and because it directly attacks the bottleneck.
- The optimal configuration is demand-dependent: (2, 4) at low to normal demand, (2, 5) at high demand, and (3, 5) only at peak, which barely satisfies a reasonable service-level objective.

Limitations and possible extensions. The model assumes exponential service times throughout, which is the standard reference but does not match every empirical distribution. Customer balking (refusing to enter the queue) and reneging (leaving after some wait) are not modelled, although the data from the high-load scenarios suggest these effects would dominate any further refinement. Extensions could include heterogeneous waiters (with different service rates), a time-varying arrival rate (peak/off-peak hours), and the inclusion of a take-away service that does not need a seat.

References

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